Belief sharing: the standard aggregation corner I

`belief_sharing.share_round` lifts the aggregation rule of sec. 5.8 to a colony of categorical sentinel agents. Each agent has a private sensory outcome and a local posterior over the same shared latent location; it broadcasts the posterior, not its raw observation. Following the sensory attenuation that the active-inference formulation of belief sharing imposes (Friston et al. 2024) — “agents do not hear themselves” — an agent’s heard consensus excludes its own message:

$$q_{-n} = \mathrm{normalize}(q/t_n)\,, \tag{1}$$

so the round in eq. 1 implements the declared categorical colony-hive-mind mechanism. With the naive fusion rule of eq. 6, eq. 1 is the project's standard log-linear-pool consensus. Under the explicit shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 5.8, it specializes Eq. 7's message-combination term; it does not reconstruct the complete source protocol. With the server-side robust rule it yields a hive-mind that can down-weight a contaminated sentinel — an effect the contamination sweep of sec. 7.5 measures rather than assumes, and one that carries no guarantee beyond the recovery limit. The per-round diagnostics — the post-sharing belief matrix, the global consensus, and the mean surprise and accuracy against a known ground-truth state — are returned by `share_round` and drive

Studies 1 and 4.

fig. 3 makes this concrete: three sentinel cards begin with different private categorical views of the nine-cell world, convert those views into local posteriors, and send only those posteriors to a fusion route. The return path is a cavity message, so the consensus heard by agent n excludes the local posterior q_n. The figure remains a protocol map rather than a new result; the empirical belief matrix and free-energy comparison remain the evidence surfaces in fig. 9 and fig. 8.

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Because eq. 1 calls the aggregation rule of eq. 6 or its robust generalization, the recovery identity of eq. 7 propagates upward: a colony running `share_round` at zero server robustness is bit-identical to a colony running the project's standard log-linear-pool round. Under the qualified categorical bridge, the pool realizes only the source message-combination specialization; the robust round is a project extension, not a reconstruction of the active-inference ensemble literature (Friston et al. 2024; Heins et al. 2024). sec. 7.2 reports that communicating colonies reach a mean variational free energy of 13.2190 nats against 16.5298 nats for incommunicado colonies across 480 seeds, with the per-agent belief matrix before and after a round shown in fig. 9 and the colony comparison in fig. 8.

The honesty boundary of sec. 3.3 carries through the lift unchanged. The robustness that eq. 1 inherits when the colony fuses with the server-side heuristic is the divergence-reweighting device of sec. 5.8, whose positive property is the naive-recovery limit and whose declared separable objective class has a scoped no-go result; the per-agent FedGVI bounded-influence result enters the colony only through the rcce/AR client losses of sec. 5.7, under the source theorem's matching assumptions, applied inside each agent's local generalized-Bayes update. The robustness sweep in sec. 7.5 and the variational supplement in sec. 11 keep the three axes distinct. The federation transport (sec. 12.4) realizes this sharing over queue-backed worker channels; by Proposition 12, the federation bit-identity result, the consensus is bit-identical to the in-process call, so the channel adds no precision loss while leaving multi-machine network transport as future work.